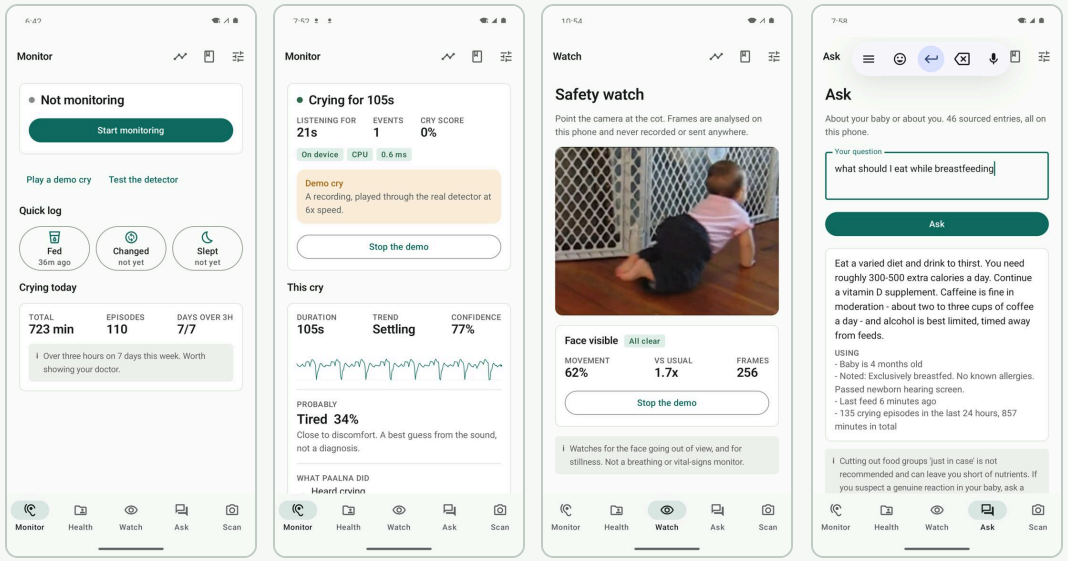
Nila

It listens, tries to settle her, and only then wakes you — on your wrist, your partner's phone, or the laptop in the next room. **Every model runs on the handset.**

Team **Just Us** · Vishal A S · Nandini K

8 models on device · 0 network calls · no subscription · works in airplane mode



Run it now, in your browser
appetize.io/app/b_u2i6ovj4ndu777nons7aebu4ku

Source code
github.com/nandini-kuppala/Nila

Two-minute demo video
nandini-kuppala.github.io/Nila

OUTLINE

What follows, in five parts.

01
The problem

03 India, in numbers
04 What the market sells

Who is suffering, how many, and why the existing answer does not reach them.

02
The product

05 The escalation ladder
06 One baby, every screen
07 Eight models on device

What Nila does, who it alerts, and what it carries to do it.

03
On the phone

08 Every inference, local
09 System architecture
10 Overall pipeline
11 Inside the audio path

The phone-first claim, drawn out step by step and measured.

04
The intelligence

12 Three pipelines
13 The cry model

Retrieval, medicine safety, vision — and the model at the centre of it.

05
Why it wins

14 Cost to the family
15 Privacy by design
16 Impact & roadmap

Zero rupees, zero network, and where this goes next.

The first 1000 days decide 90% of a child's brain development. India spends them under-supported.



36%

of children under five are **stunted** — and over a third suffer long-term malnutrition

NFHS-5, 2019–21



26%

of mothers in nuclear families report **severe loneliness** — no one to ask at 3 a.m.

ScienceDirect, 2024



20%

of infants **cry excessively** in the first months — the peak is six to eight weeks

Jornal de Pediatria



19%

of Indian mothers experience **postpartum depression** — 22% across the south

Meta-analysis, 59 studies

INFANT MORTALITY · 2024

25.8

per 1,000 live births

BRAIN DEVELOPMENT

90%

inside the first 1000 days

CRYING, PER DAY

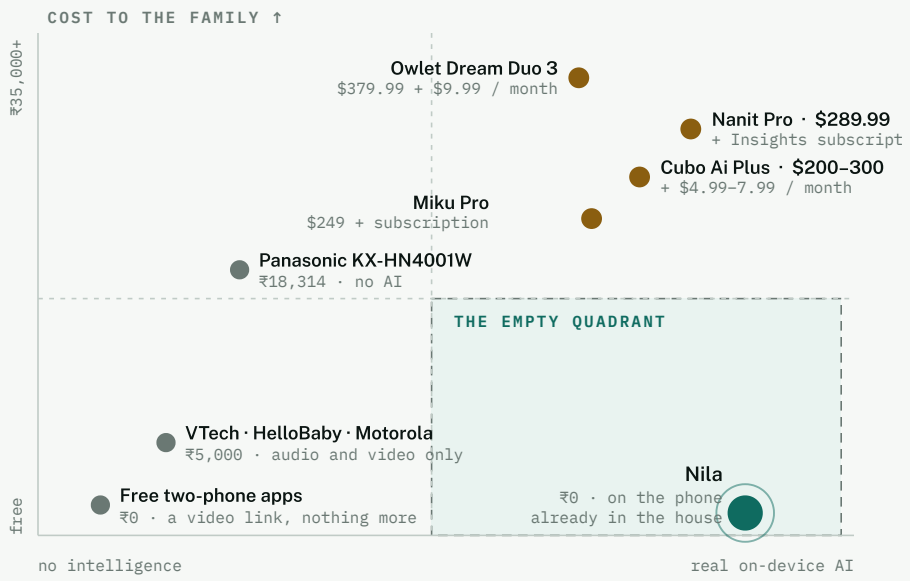
2–3 HRS

normal for a young infant

WHAT THAT ADDS UP TO, AT 3 A.M.

- **Normal crying is the most common trigger** for caregiver crisis — and it arrives exactly when the household is most exhausted.
- **Nobody is awake to ask.** A quarter of mothers have no one in the house to turn to, and the questions are urgent: is this normal, is this medicine safe while I am feeding her, has she fed enough?

Real intelligence is locked inside expensive hardware.

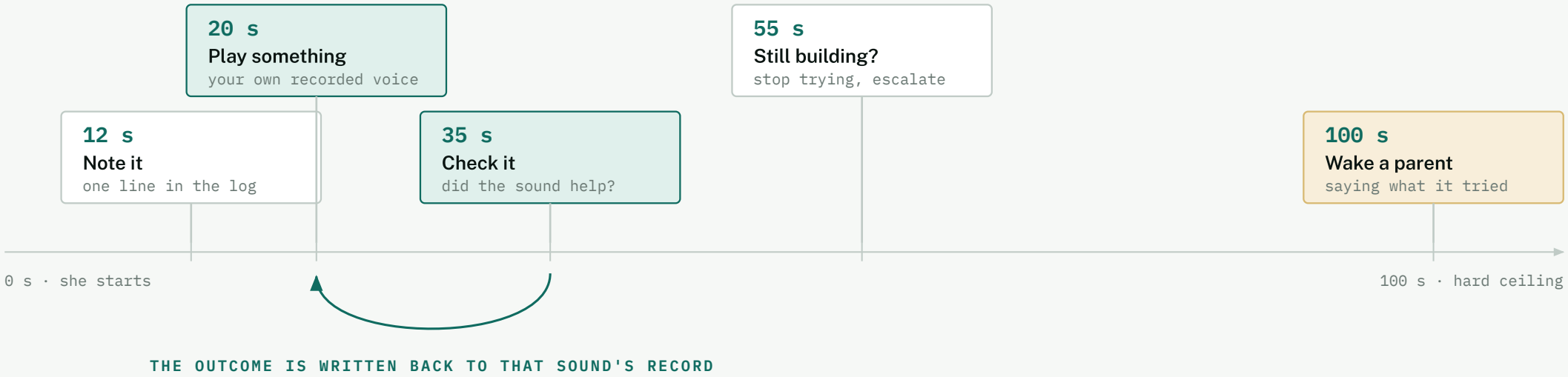


Prices as listed by each vendor, 2026. Indian street prices for imported units run higher still.

PRODUCT	HARDWARE	ONGOING	WHERE THE AI RUNS
Owlet Dream Duo 3	$379.99	$9.99/mo	Vendor cloud
Nanit Pro	$289.99	Insights plan	Vendor cloud
Cubo Ai Plus	$200–300	$4.99–7.99/mo	Vendor cloud
Miku Pro	$249	Subscription	Vendor cloud
Panasonic KX-HN4001W	₹18,314	–	No AI
HelloBaby HB6339	₹5,131	–	No AI
Nila	**₹0**	**₹0**	**On the handset**

- **Every intelligent monitor sends the nursery to a server.** 1.1 million Wi-Fi baby monitors were reachable across 118 countries on one extracted key.
- **All of them stop at *notify*.** None of them try to settle the baby, and none check whether what they tried worked.

It tries to settle her first — then checks whether that worked.



Real constants from the shipped build. The ladder is a loop, not a line.

Nothing happens first

A fuss that resolves itself is not an event. The first twelve seconds are logged silently.

Your voice goes first

Recorded lullabies outrank every built-in sound. Volume is capped at 55% — this plays a metre from an infant's ear.

It measures, not assumes

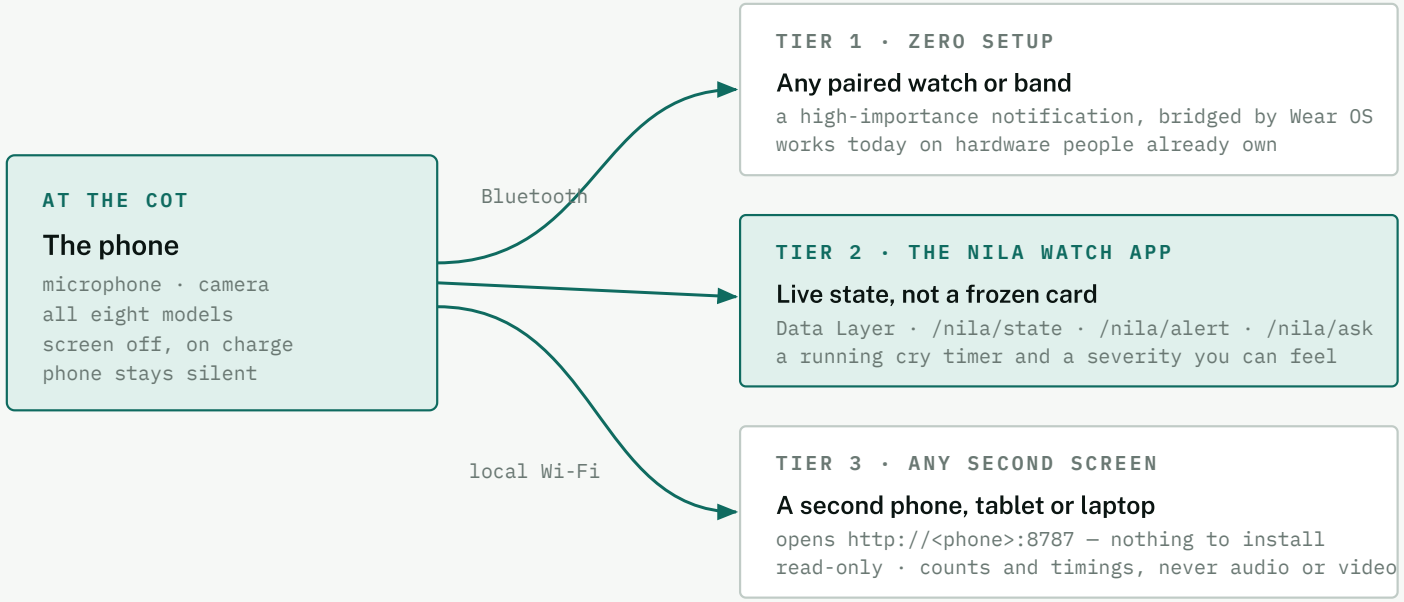
Settled means a settling trend or an 18 dB drop from peak. The answer is recorded either way.

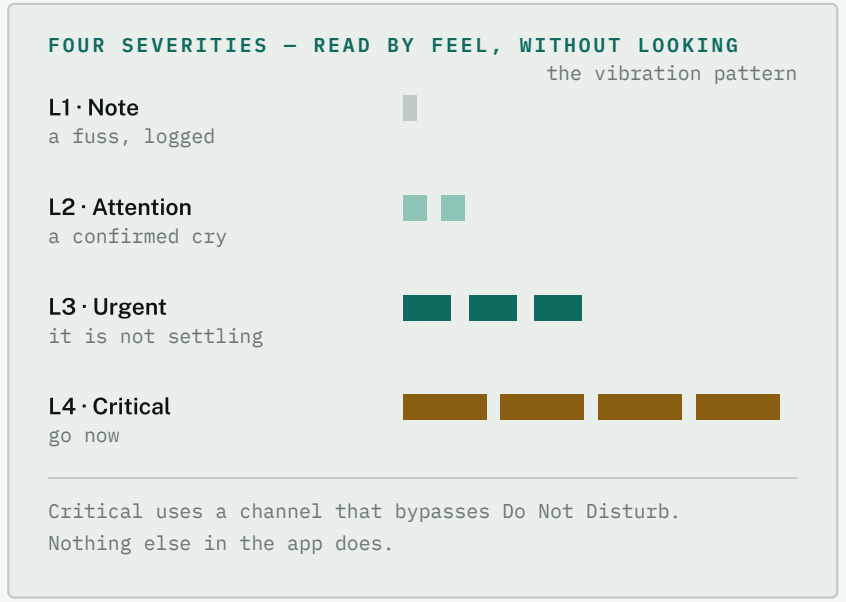
It learns this baby

A smoothed success rate per sound, with 15% exploration so the first thing that worked is not played forever.

The verification step is the product. No monitor on the market attempts to settle the baby, and none of them check.

The phone watches the cot. The alert finds you wherever you are.





Every push is best-effort and fails silently on purpose — the phone's own alarm is the failsafe. A safety alert must never depend on a radio link.

Stepped out to the shop?

The watch buzzes on your wrist with the severity in the pattern, and the alert names what Nila already tried.

Working in the next room?

Open the dashboard on a second phone or the laptop. Live status, the night's timeline, no app to install.

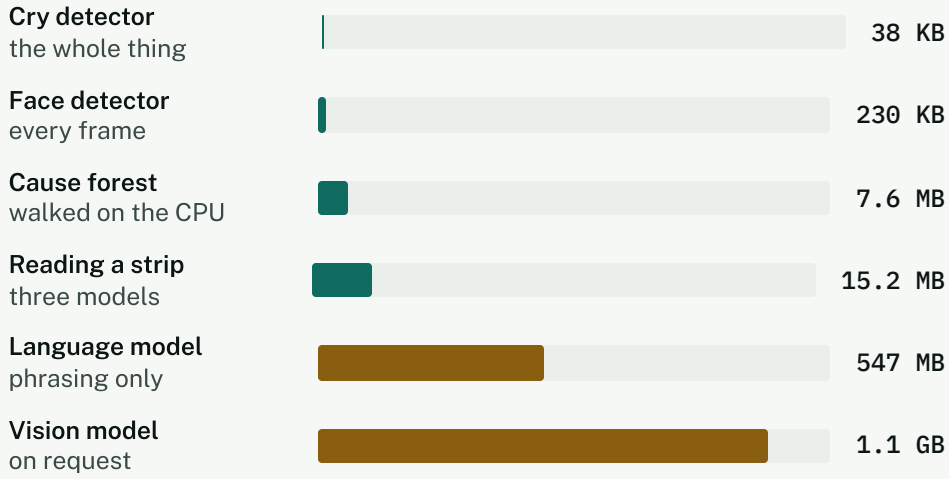
Both parents, one baby

Every paired device receives the same state at the same time. There is no primary account and nothing to log into.

Eight models. One phone. Zero servers.

MODEL	PLAIN ENGLISH	SIZE
Cry detector · CNN	Is that a baby crying, or the washing machine?	38 KB
Cause forest · 500 trees	A first guess at why — hungry, tired, discomfort	7.6 MB
PP-OCRv4 detection · DBNet	Finds the text on a strip, even at an angle	15.2 MB
PP-OCRv4 angle	Works out which way up the packet was	
PP-OCRv4 reading · CTC	Turns those pixels into 6,625 possible characters	
BlazeFace	Can the camera still see her face?	230 KB
Qwen2.5-0.5B	Shortens an answer retrieval already found	547 MB
FastVLM-0.5B	Describes the cot when asked for a second opinion	1.1 GB

RELATIVE FOOTPRINT, ON DISK



THE LISTENING CORE

268 KB

detector and face model together — smaller than one photograph

SETUP STEPS

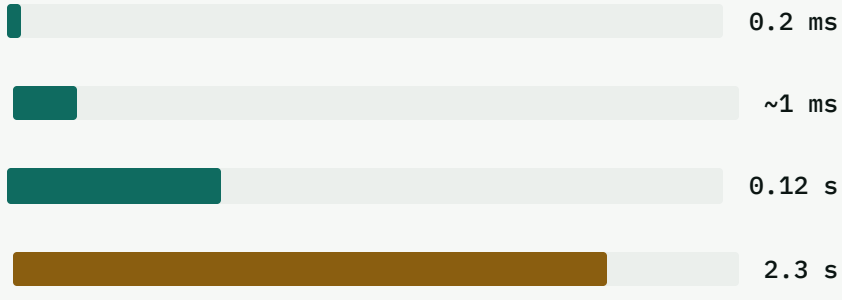
0

install the APK and every feature is live

Plus the knowledge base — **46 sourced entries** from NHS, WHO, AAP, IAP and LactMed, 45 KB, and a local database holding the care log, health records and soother outcomes.

Every inference runs on your iQOO.

MEASURED, PER INFERENCE



TRIED IN THIS ORDER, EACH VERIFIED WITH A REAL INFERENCE



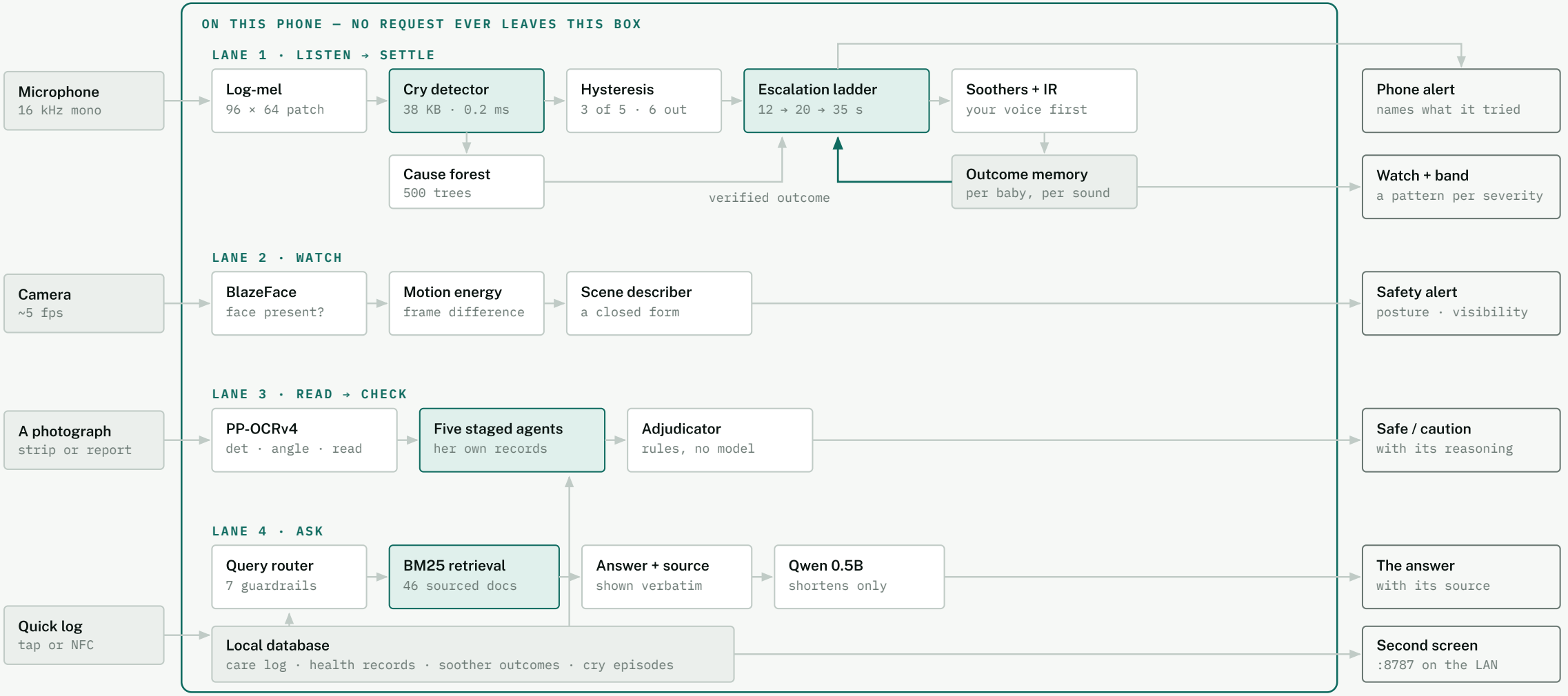
WHAT THAT BUYS, OVERNIGHT

- **A whole night on one battery.** 206 MB resident while monitoring, about 0.04% of a single core — because below −55 dBFS the pipeline skips inference entirely, which in a quiet nursery is nearly every window.
- **Never a surprise at 3 a.m.** A delegate that builds successfully can still fail on its first real frame, so each one is probed before it is trusted.
- **Provable on stage.** One tap runs the real audio path on the real accelerator and prints its confidence, per-window latency and compute unit.

NETWORK CALLS	MEMORY, MONITORING
0 cleartext is forbidden outright in the manifest	**206 MB** the state it holds for eight hours
ONE CPU CORE	**SECOND SCREEN**
0.04% one window every 480 ms	**8787** the dashboard port, on the local network

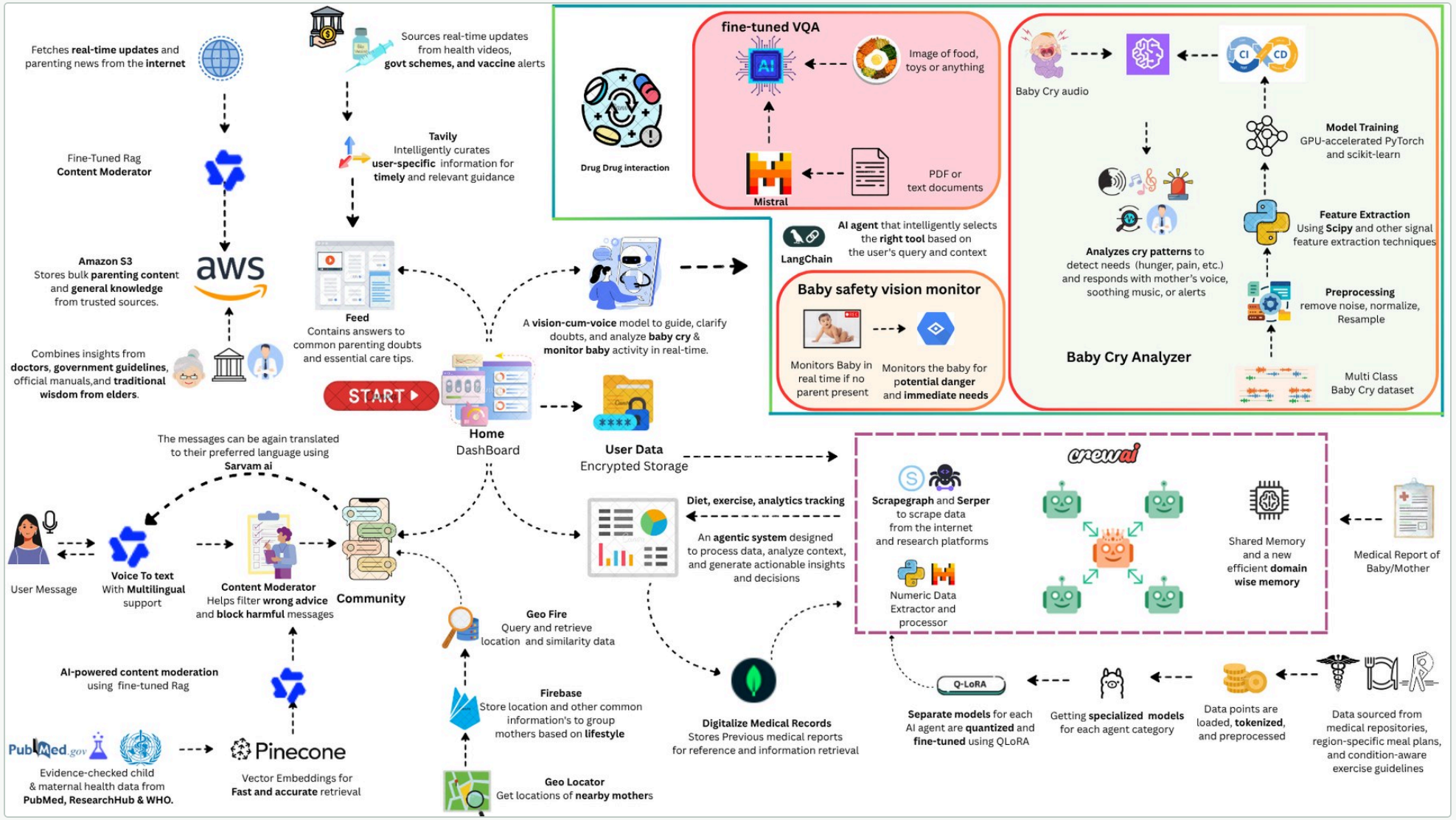
iQOO 15 · Snapdragon 8 Elite Gen 5 · 80 TOPS Hexagon NPU · arm64-v8a · minSdk 26 · TensorFlow Lite, ONNX Runtime and MediaPipe, all on device

Four lanes, one boundary, nothing crosses it.



Solid teal = the paths that matter most. Everything drawn here executes on the handset; the only thing designed to leave is a one-page PDF for a paediatrician.

Overall pipeline



From a microphone buffer to a decision, ten times a second.



Why no auto gain control: it flattens the loudness envelope, and the envelope is exactly what the ladder reads to decide whether a sound helped.
Why 50% overlap: a cry starting mid-window would otherwise be split across two patches and score weakly in both — that onset is what matters most.

Retrieval, medicine safety, and vision — each one staged.

ASKING A QUESTION · OFFLINE RAG

46 sourced documents

Route it first
emergency · dose · diagnosis · distress · too vague · off topic

Retrieve
BM25 keyword ranking over 46 docs · title ×4 · keywords ×3

Filter
by audience and the baby's age in months

Answer
the retrieved text, verbatim, with its source

Shorten, optionally
Qwen 0.5B · checked for overlap, quantities and polarity

CHECKING A MEDICINE · FIVE AGENTS

98% accuracy · drug–drug interaction

1 · History
her conditions, mined from her own records

2 · Read
PP-OCRv4 · rotated, upside down, white on foil

3 · Identify
exact tokens, then fuzzy — 1BUPROFENA00 → ibuprofen

4 · Interact
against her allergies, conditions and current medicines

5 · Adjudicate
rules only · the worst finding always wins

WATCHING THE COT · TWO SIGNALS

every frame, all night

Always on
BlazeFace presence + frame-difference motion, every frame

Something changed
the face is gone — but which of the several meanings is it?

One frame, one question
POSITION · FACE · CONCERN · DOING — a closed vocabulary

Parse strictly
anything outside the vocabulary is discarded, not shown

Alert only on concern
uncertainty is never an alert · frames never touch disk

One rule holds across all three. A language model never supplies a fact, is never shown a health record, and is nowhere near the medicine verdict — it phrases what retrieval already found, and its output is dropped if it drifts.

Drug–drug interaction accuracy measured on the DrugBank dataset.

The fastest and most accurate baby cry model.

MACRO AUC	RECALL ON CRYING
0.992	**98%**
on infants never heard in training	it catches what it exists to catch
PER WINDOW	**ON DEVICE**
0.2 MS	**38 KB**
one window every 480 ms	21,218 parameters, quantised

HOUSEHOLD SOUNDS IT DOES NOT FIRE ON

vacuum cleaner · washing machine · door knock · can opening · coughing · cat · siren · car horn · crying baby · 8 of 8

WHY IT IS THIS GOOD

- **Hard-negative mining.** Every training negative was scored with the trained model and the hardest 18% oversampled — cats, laughter, roosters, crows, toilet flush. Cry F1 rose from 0.922 to 0.942 and the household false alarms went to zero.
- **A frontend that is a known quantity.** 25 ms frames, 10 ms hop, 64 mel bands from 125 to 7500 Hz — and infant cry fundamentals sit at 300–600 Hz, right in the middle of it.
- **Written twice and pinned together.** The transform exists in Python for training and Kotlin for inference; a golden fixture holds them within 0.002, so the phone sees exactly the distribution the model was trained on.
- **Quantised, then re-checked.** Compressing to 38 KB moved the AUC by 0.00006 — which is why the small model is the one that ships.

Tested on the Donate-a-Cry corpus · subject-wise held-out infants, no infant in both halves · 2,946 test clips

No second device. No subscription. No cloud bill.

What the market asks for

- × **$249 – $379** for the camera, before anything else
- × **$5 – $10 every month**, or the intelligent features switch off
- × **Nursery video on a vendor's server**, in another country
- × **A working network**, or it stops being a monitor
- × **One more device** to buy, mount, charge and replace

What Nila asks for

- ✓ **₹0** — it runs on a phone the household already owns
- ✓ **No account and no subscription**, now or ever
- ✓ **Nothing leaves the device**, because nothing is ever sent
- ✓ **Works with the router unplugged**, in airplane mode, on a train
- ✓ **Install and it works** — no pairing, no setup, no download

WHY AN IQOO MAKES IT BETTER

- **The Snapdragon NPU** keeps detection instant with the screen off and the phone on charge beside the cot.
- **An infrared emitter.** Nila turns the fan or AC down at the first soothing attempt. Very few phones ship one — iQOO does, and Nila speaks NEC frames to it.

THE MARKET IT OPENS

- **Asia-Pacific is the fastest-growing baby-monitor market** in the world at roughly 10.9% a year, and India is most of that growth.
- **India has 700+ million smartphones** and a spare handset in a great many drawers. That is the installed base Nila ships to on day one.

WHO IT REACHES THAT HARDWARE CANNOT

- **Homes with no broadband,** where a cloud monitor is simply not a product.
- **Families for whom ₹25,000 is not a baby purchase** — which is most of them.

Private by construction, not by policy.

No network requests

Cleartext is forbidden outright in the manifest. The app holds no API keys, because it calls no APIs — there is nothing in the build to extract.

No audio is recorded

Monitored sound is analysed in memory and discarded, window by window. The lullabies you record are the one exception, and they stay on the handset.

Backup is disabled

Turned off in the manifest, so health documents for the baby and the mother cannot leave by a cloud transport even by accident.

No account, no analytics

Nothing to sign up for and nothing measured about you. The second screen is off until you switch it on, and refuses anything but a local address.

SEVEN GUARDRAILS, ALL OF THEM LOCAL

Emergency	20+ phrasings — not breathing, blue lips, seizure, choking
Dose requests	Always refused. The build fails if a dose appears in the corpus
Diagnosis	Declines, and says what to observe and when to seek care
Caregiver distress	Safe immediate action first, then helpline numbers
Age gating	Newborn guidance never returns for a nine-month-old
Audience routing	Her questions reach the 27 documents written for her
Relevance floor	Says it has nothing rather than offering the nearest paragraph

A DEMO THAT WORKS ANYWHERE

- **It seeds itself with an example family** — a four-month-old, her mother's record with a penicillin allergy, a day of feeds and a week of crying — so no screen is ever empty on a device nobody has used.
- **Every time in it is relative.** "165 minutes ago", never a date, so a build made today still shows a baby who fed two hours ago next month.
- **It runs on a plane, in a browser and in a hall with no signal**, which is exactly where this is being judged.

Nila is an awareness aid. It reports what a microphone and a camera can establish — crying, its duration, its direction, whether a face is visible and whether there is movement. Knowing precisely where that line sits is what makes everything on the right side of it trustworthy.

A monitor that settles the baby, on a phone already in the house.

WHAT CHANGES

- **The household stops being woken unnecessarily.** Two rungs run before any alert, so a baby who would have self-settled does — and the parent who is woken is woken for a reason, with that reason attached.
- **The alert follows the parent.** Wrist, second phone or laptop, with the severity readable by feel — so nobody has to sit in the room to be on duty.
- **The mother is a user, not a bystander.** 27 of the 46 entries are hers: diet, lactation, recovery, mental health, and twenty medicine monographs checked against her own allergies and conditions.
- **It works where the network does not** — a village, a train, a basement ward, a power cut. The app cannot tell the difference, because it never asks.

WHAT COMES NEXT

- **On-device personalisation.** A few minutes of one specific baby's cry, fine-tuning the final layer on the handset itself — learning that never leaves the phone, which is the natural next step for a device this capable.
- **Ten Indian languages.** The language field already runs through the profile, router and retriever; the corpus is next, with human review.
- **A 200-entry knowledge base**, widening from crying and medicines into night weaning, reflux, stool changes and growth.
- **A shared night.** Two caregivers, one baby, alternating who is on call — the device mesh already carries everything that needs.

Try it — a real Android device, in your browser
`Monitor → Play a demo cry · Watch → Play demo footage`

Source, models and every evaluation report
`github.com/nandini-kuppala/Nila`

Technical record · eighteen sections
`nandini-kuppala.github.io/Nila/technical-record.html`

Sources

OFFICIAL STATISTICS

- Press Information Bureau, Government of India — infant and child health releases, 2023 and 2024
- World Bank — mortality rate, infant, India, 2024
- National Family Health Survey (NFHS-5), 2019–21 — stunting and child nutrition
- Registrar General of India — Sample Registration System bulletins
- National Health Mission — Child Health programme data

CLINICAL AND ACADEMIC

- Prevalence of postpartum depression at subnational level in India — systematic review and meta-analysis
- Goswami, I. et al., ScienceDirect 2024 — maternal loneliness in nuclear families
- Pediatric Abusive Head Trauma — StatPearls, NCBI Bookshelf
- Excessive crying in infants — Jornal de Pediatria
- Wessel criteria and Rome IV — infantile colic
- NHS · WHO · AAP · IAP · LactMed — the 46 entries in the knowledge base, each carrying its own source

TECHNICAL AND MARKET

- ESC-50 and UrbanSound8K — environmental negatives · Donate-a-Cry corpus — 457 clips from 221 infants
- PP-OCRv4 and Qwen2.5-0.5B-Instruct (Apache-2.0) · FastVLM-0.5B (Apple ML Research)
- MediaPipe, TensorFlow Lite and ONNX Runtime (Apache-2.0)
- Qualcomm — Snapdragon 8 Elite Gen 5 Hexagon NPU platform brief
- Consumer Reports — Wi-Fi baby monitor exposure disclosure, 1.1 million units
- Vendor pricing as published by Owlet, Nanit, Cubo Ai, Miku, Panasonic and HelloBaby, 2026

Code is MIT licensed. The cry detector inherits ESC-50's non-commercial terms. Nila is an awareness aid, not a medical device: it does not monitor breathing or vital signs, never states a dose and never diagnoses.